

Cellular and Medical Immunology Defense Against Infectious Agents and Vaccines

Integrated Biomedical and Clinical Sciences OMFS 513

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Learning Objectives

- 1- Describe integrated response to infection.
- 2- Compare and contrast active and passive immunity.
- 3- Describe the immune response to vaccination.
- 4- Describe different types, adv and disadv. of vaccine strategies.
- 5- Describe common side effects associated with vaccine administration.

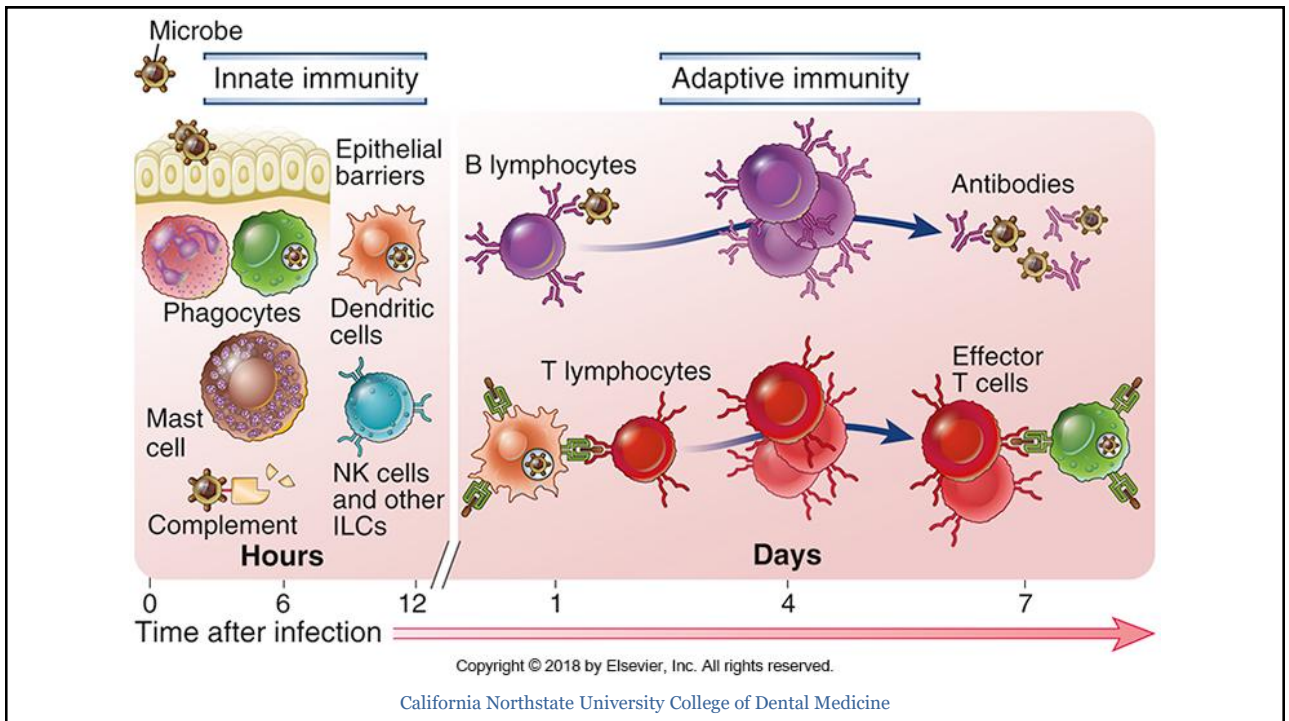
Outline

1. Integrated response to infections
2. Historical aspect for vaccines
3. Active and passive immunization and herd immunity.
4. Immune response to vaccines.
5. Adjuvant
6. Types of vaccines.

Integrated Immune response

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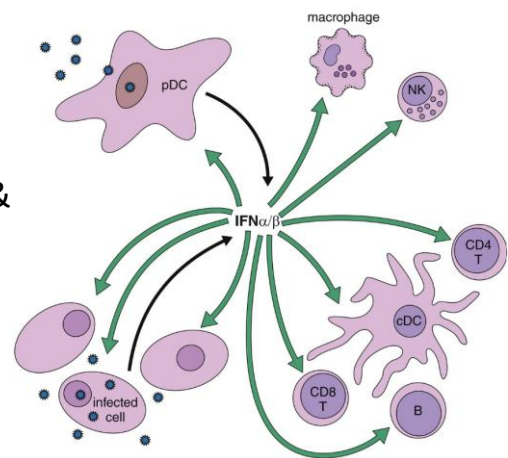
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Host Defense Against Viruses

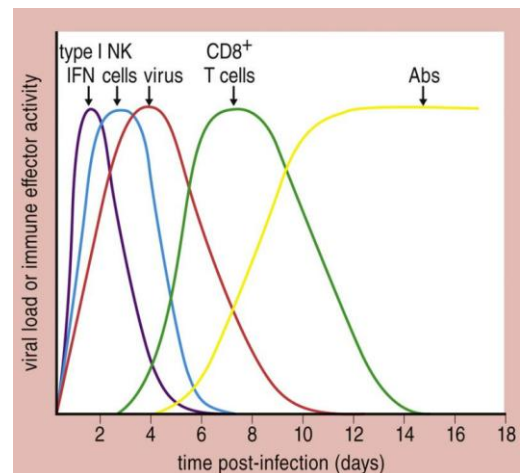
1) Innate immune response

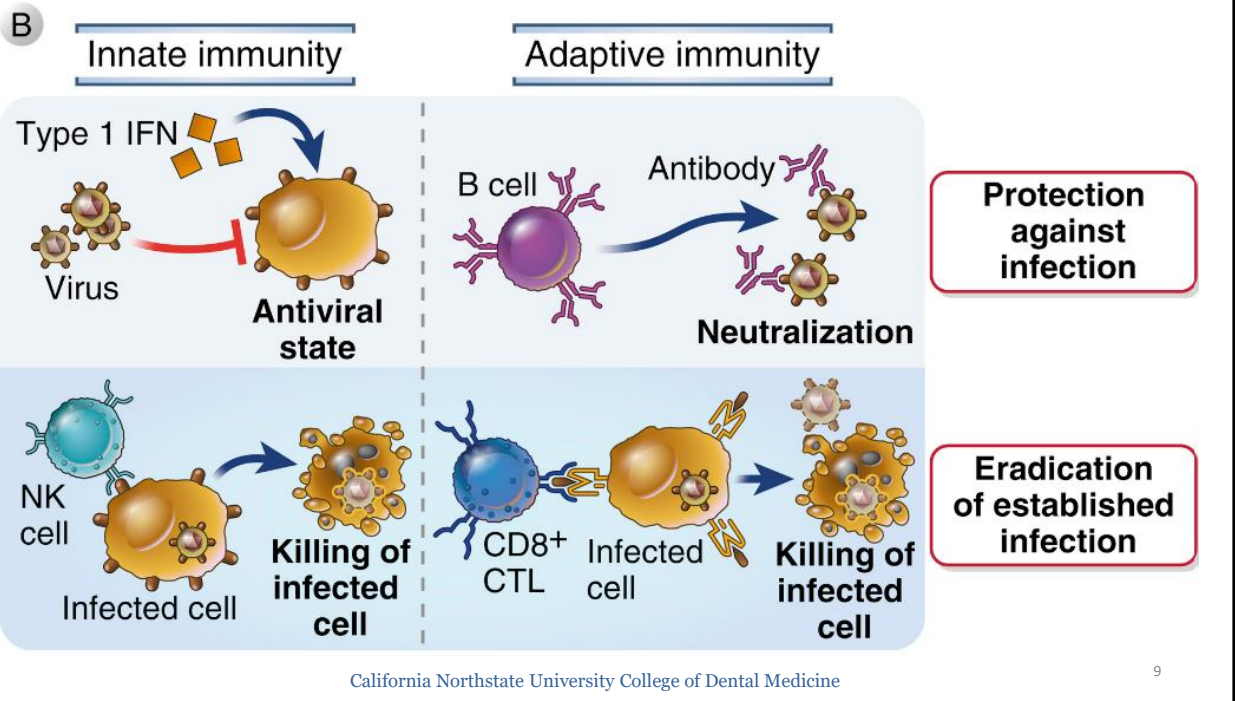
- a) Barriers.
- b) Inflammation, activate DCs and macrophages.
- c) Inhibition of infection by interferons (IFN) & TNF- α . IFNs upregulate the expression of IFN-stimulated genes (ISGs). Antiviral ISGs inhibit virus replication.
- d) Activated NK cells are detected within 2 days of viral infection. NK cells recognize and kill virus-infected cells. Example, NK cells play a key role in herpesvirus control.



2) Adaptive immune response

- a) Cell-mediated immunity. cytotoxic T lymphocytes and cytokine release from T-helper cells
- b) Humoral immunity. B lymphocytes produce neutralizing and non-neutralizing antibodies against viral-specific antigens (function?).





Extracellular Bacteria

Innate immune response:

Barriers. Recognition.

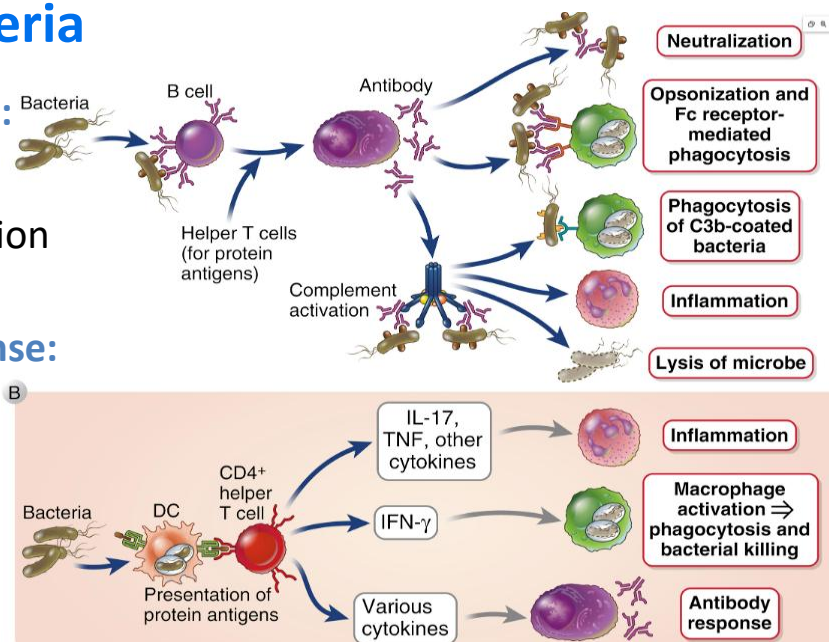
Complement. Inflammation

Phagocytes.

Adaptive immune response:

B-cell humoral.

T-cell (CD4 cells secrete IL-17, TNF and IFN γ).



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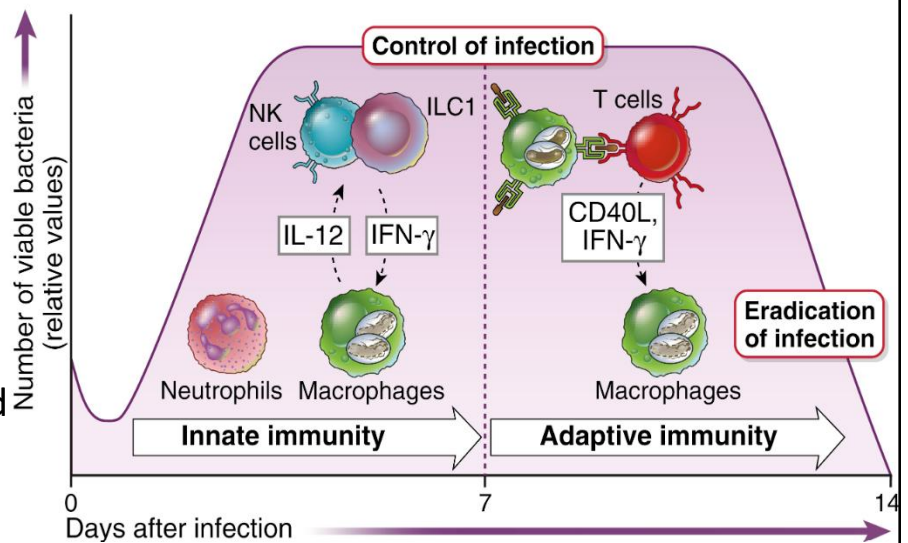
Intracellular Bacteria

Innate

NK cells, neutrophils and macrophages.

Adaptive

Mostly cell mediated (CD4 and CD8 cells).



Inflammatory response

Bacteria can cause two types of inflammation reaction

1) Pyogenic:

In pyogenic (pus-producing) inflammation, neutrophils are the predominant cells.

2) Granulomatous inflammation:

Macrophages and helper T cells predominate. The most important organism in this category is *Mycobacterium tuberculosis*.

Immune Response to Fungal Infection (candidiasis)

No approved vaccines. Efficient host immunity is essential for prevention.
Infections more common in immune compromised.

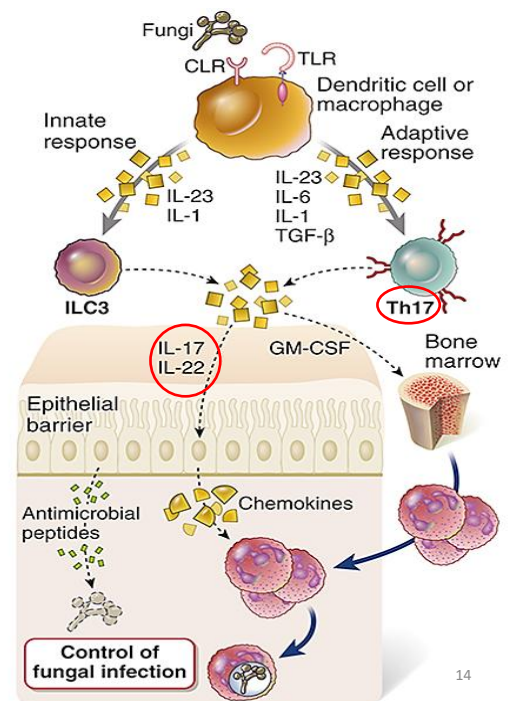
A) Innate immunity:

- 1- Antimicrobial peptides
- 2- Recognition: C-type lectin receptors (CLR) & TLR2, release inflammatory mediators, activate neutrophils and macrophages.
- 3- Complement system: defense for systemic infection. Opsonization is the main outcome.

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B) Adaptive immunity:

- 1- Cell-mediated. Extracellular fungi elicit strong Th17 responses that recruit neutrophils. Intracellular fungal infections can promote Th1 response which initiate granulomatous inflammation.
- 2- Humoral response (IgA in the oral mucosa) and serum ab for systemic.



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Immune Memory is a Unique Feature of Adaptive Immunity

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Protection via Vaccines

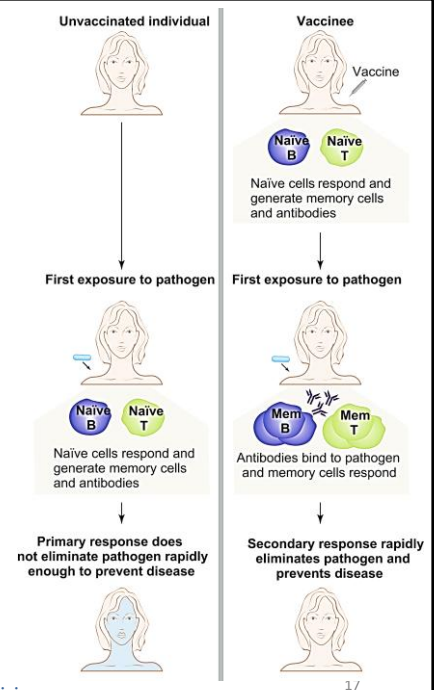
A prophylactic vaccine is a biological product used to safely induce an immune response to confer protection against an infection on subsequent exposure to a pathogen.

Some vaccines are combined with an adjuvant to improve their ability to induce an immune response (**immunogenicity**) by providing danger signals to the innate immune system.

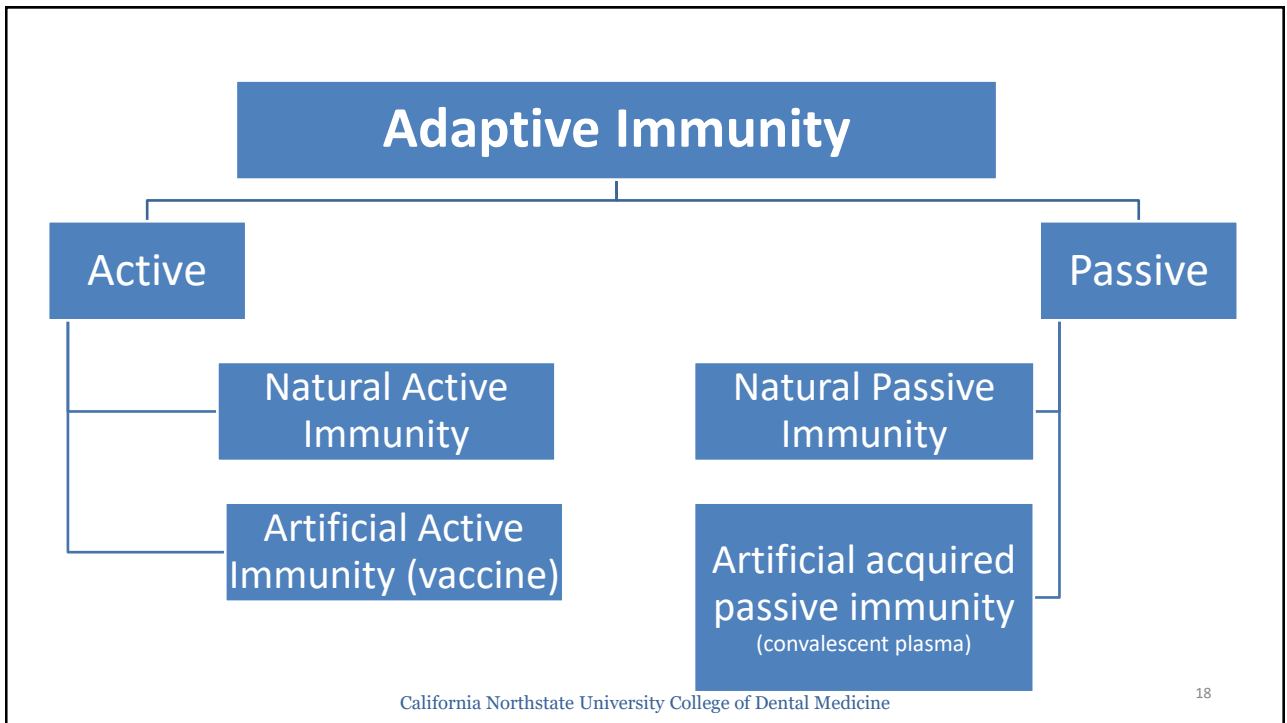
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Natural infection: the body mounts innate and adaptive immune response to create memory cells. Usually, accompanied by disease symptoms.

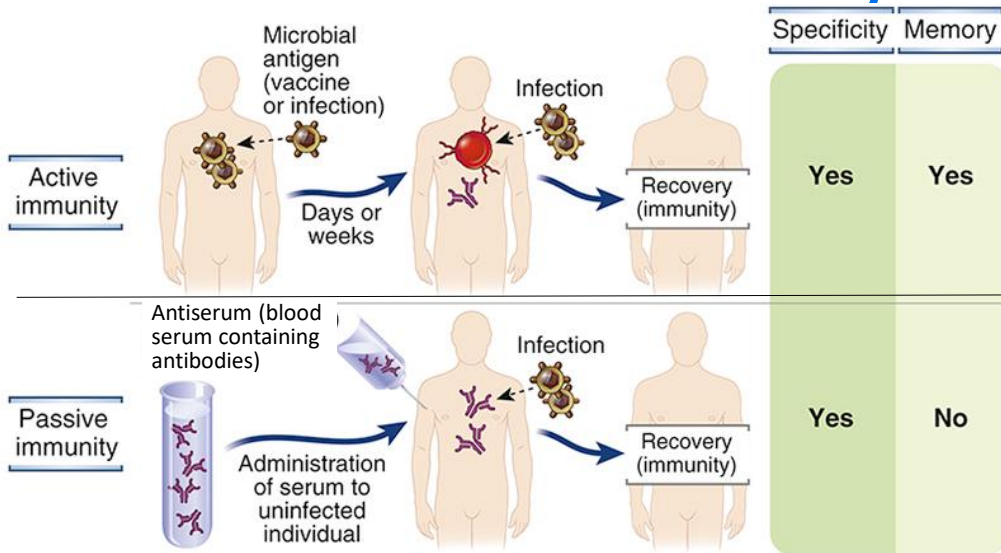
Vaccines does not cause disease. Produce an immune response that provides long-lived protection (memory B and T-cells) against subsequent infection.



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Active and Passive immunity



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History of Vaccines

- Smallpox is caused by *Variola major*.
- Smallpox caused an epidemic in North America in 1775–1782.
- Smallpox caused 30 % mortality rate and survivors had disfiguring scars.



SMALLPOX THREAT	
WHAT IS SMALLPOX?	EFFECTS ON HUMAN BODY
? - Airborne virus, highly contagious - Caused by variola virus - Virus related to monkeypox - Could take approximately 6 weeks to seed smallpox cases around the world - Each infected individual infects an average of 20 others Early symptoms High fever, fatigue and rash Resulting spots fill with clear fluid and pus then form a crust and fall off Fatality rate ranges from 15 to 50 percent Treatment Vaccination (vaccinia) to prevent infection - During the first four days after exposure - Current U.S vaccine stock-pile was produced in 1970s and contains 7.5 million doses - To be kept on shelves and used in confirmed cases only	Spreads through saliva droplets in an infected person's breath 1 2 Incubation period of 7-17 days after exposure Infectious once rash develops and first week of illness HISTORY 1976 WHO urges labs worldwide to destroy retained stocks of variola virus 1977 Last known natural case - Somalia 1978 Laboratory accident in Birmingham, England, kills one and causes a limited outbreak 1983 South Africa - last country to destroy its virus stock

Source: WHO

REUTERS

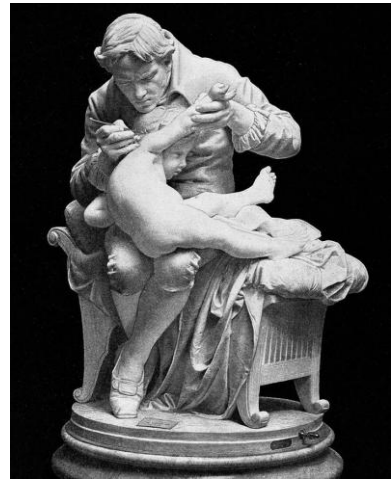
Nigeria Natural Health Online - Africa's foremost blog on natural health:
 Red Alert: Smallpox May Soon Come Back! - Experts (kimekwu.com)

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Cowpox can confer protection for Smallpox

- Cowpox, related zoonotic virus, caused a milder disease than smallpox.
- In 1796, Dr. Edward Jenner noticed that milkmaids who contracted Cowpox were immune to Smallpox infections.
- Exp. with James Phipps (8Y) “Cowpox infection can protect against Smallpox infection (vaccine development)



Edward Jenner, memorialised here in a statue vaccinating his son, swerved ethical concerns to see whether a cowpox vaccine could save lives (Credit: Getty Images)

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- The word vaccine (from Latin word Vacca or cow) was first proposed by Louis Pasteur.
- Pasteur observed that weakened or attenuated pathogen strain could be administered to provide immunity against the disease.
- In 1885, Pasteur administered his first vaccine of attenuated rabies virus. Later, Pasteur institute was able to treat many rabies victims. He also vaccinated sheep with anthrax bacteria that were attenuated by heat treatment.



Louis Pasteur watching Joseph Meister receive the rabies vaccine

Strategies for Vaccine development

- Contain antigens: derived from the pathogen itself or produced synthetically to represent components of the pathogen “protective antigens”.
- Don’t cause a disease but elicits an immune response that provides protection against infection by the live, pathogenic microbe “prophylactic vaccine”.
- Long-lasting: Induce appropriate immune response (memory cells). Antibodies can neutralize microbes. T_h response protect against intracellular pathogens.
- Most effective against infections that are limited to human hosts and do not have animal reservoirs.
- Safe, tolerated, stability, cost...

Developing a good vaccine

The following microbial properties lead to an efficient vaccine:

- 1- Absence of latent infection.
- 2- Low antigenic variation (low mutation)
- 3- Microbe does not interfere with the host immune response.
- 4- Preferable: microbes that induce an acute rather than chronic diseases.

Let's apply HIV?

Challenges facing vaccines

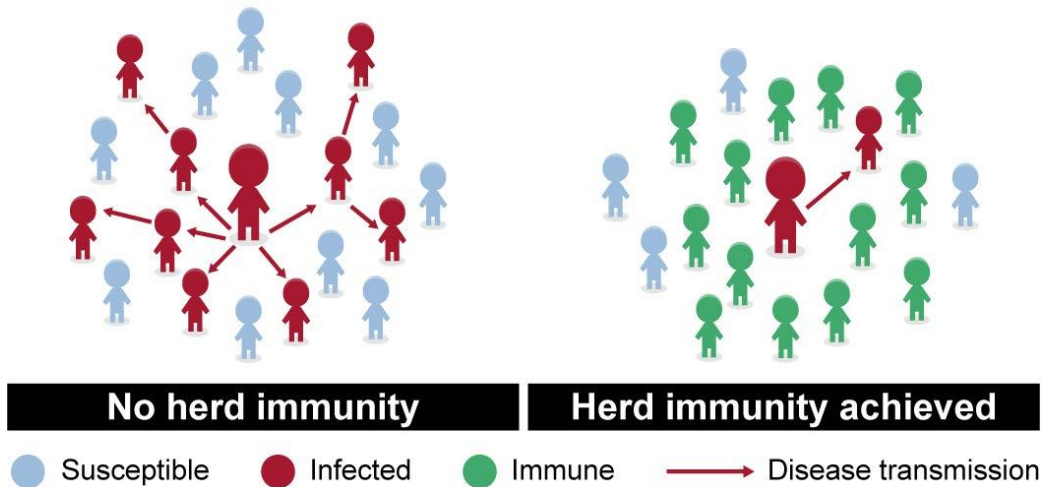
Longevity of immune response to some vaccines is variable. Reason for the differences is not well understood.

High manufacturing costs, time and investments are required for mass production, shelf-life instability and problems related to vaccine administration and vaccine hesitancy.

Development of universal vaccine is challenging for certain microbial strains as well as strains with high mutation rate.

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Aim: Herd Immunity



Source: GAO adaptation of NIH graphic. | GAO-20-646SP

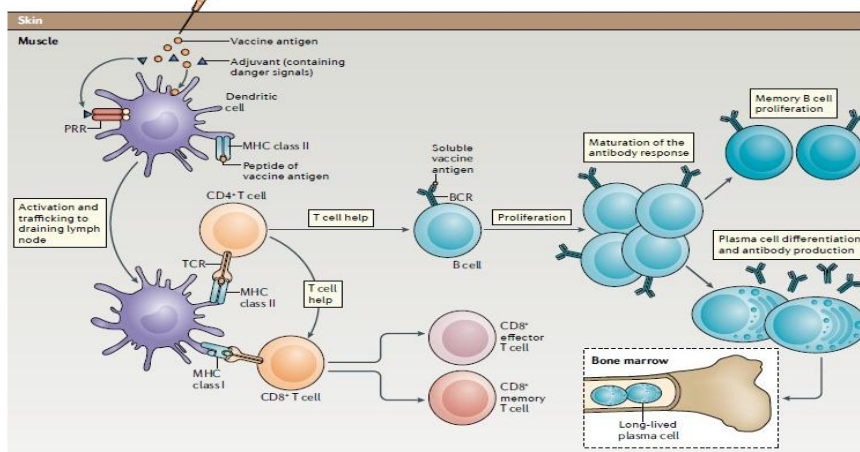
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Significance of humoral immunity

1. Antibodies fight off infections via multiple pathways.
2. Individuals with immunological defects in generating antibodies develop weak immunity following vaccination.
3. Passive protection is effective. IM or IV infusion of exogenous antibodies can provide protection against some infections.
4. Polysaccharide (subunit) vaccines: Vaccines are made from surface polysaccharides of bacteria such as meningococci (*N. meningitidis*) and pneumococci (*S. pneumoniae*). Those are T-independent antigens (no memory T-cells) but their vaccines provide considerable protection against the respective diseases.

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T-cell response



- T-cells recognize antigens on MHC and develop helper and cytotoxic cell populations as well as memory T cells.
- Helper T cells further activate the B cell response and induce class switching.
- B cells recognize the antigen & develop into plasma cells. B cells also develop memory B cells. Memory B cells will ensure faster and stronger response on future subsequent exposures.

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Timing and spacing of vaccine doses

- Most vaccines require two or more doses for an adequate immune response.
- In general, decreasing the interval between doses in a multiple-dose vaccines can interfere with the antibody response and protection. Exception: administering a vaccine dose up to 4 days before the minimum age or interval to avoid missing an opportunity to vaccinate is considered valid immunization.
- Simultaneous administration of vaccines is allowed, to increase the chances of an individual being fully vaccinated at the appropriate age (in most cases).

Factors affecting the vaccine schedule

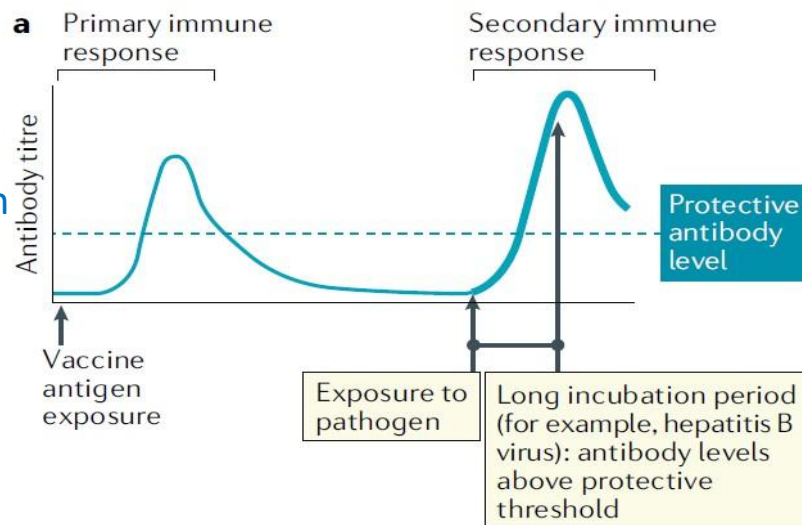
Many factors influence the number of doses recommended in a vaccine series including, type of vaccine, epidemiology of the disease, and host factors.

1- Type of vaccine

- **For live, attenuated vaccines**, the first dose administered at the recommended age usually provides protection. Additional dose can offer more protection.
- **For inactivated vaccines**, the first dose, administered at the recommended age, usually does not provide protection. Therefore, additional doses are required to mount the protective immune response.

2- Epidemiology of the disease

A. Pathogens with long incubation period



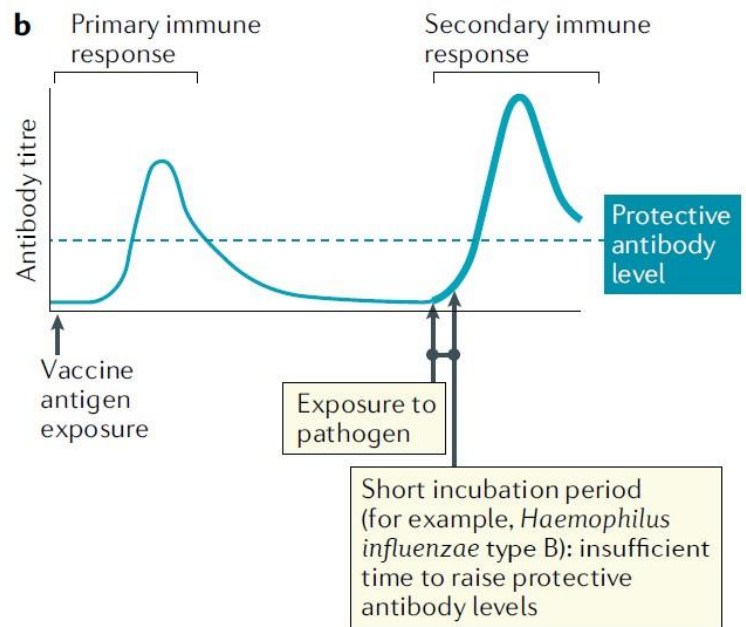
Longer incubation period allows enough time for the human body to develop a strong immune response.

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B. Pathogens with short incubation period

In this cases, immune memory may not be sufficient for protection against rapid invasive infections.

Therefore, better protection depends on repeated immunization doses. This will produce high levels of neutralizing antibodies in circulation.

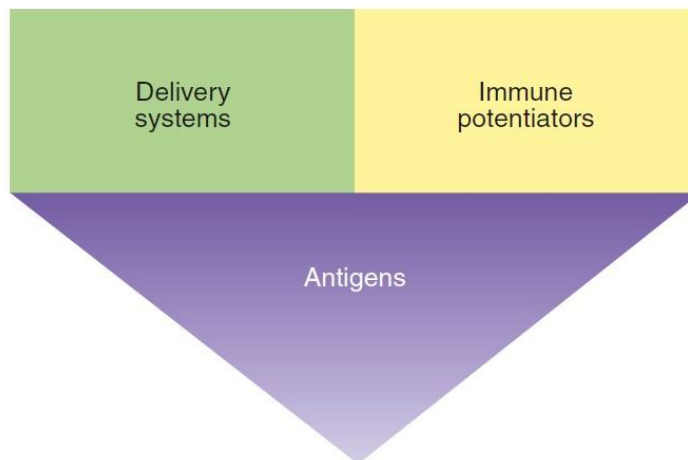


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Key components of an effective vaccine

Effective vaccines have three key components:



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Adjuvants

- Adjuvants are added compounds to enhance the innate immune response to the antigen == stronger adaptive immune response.
- At the molecular level, adjuvants induce the upregulation of co-stimulatory molecules and cytokines production. Therefore, they activate dendritic cells and other APCs to stimulate the adaptive response. Some adjuvants also work as delivery vehicles to encourage the uptake of the antigen by APCs.

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Examples of adjuvants:

Aluminum salts (alum). Alum injection induce inflammation that tends to promote Th2 responses ($>Th1$) and humoral immunity. Alum creates slow-release delivery of the immunogen at the injection site. Helps in sustained stimulation of the immune response; used in multiple vaccines.

Newer adjuvants designed to contain ligands for pathogen recognition receptor like CpG oligonucleotides to mimic bacterial and viral genetic material (for TLR9), like Hep B vaccine.

Newer adjuvants are being developed to target specific components of the body's immune response, leading to a stronger immune response.

[Adjuvants and Vaccines | Vaccine Safety | CDC](#)

[Emerging concepts in the science of vaccine adjuvants - PMC \(nih.gov\)](#)

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What else is in vaccines?

[What's in Vaccines? Ingredients and Vaccine Safety | CDC](#)
[Common Ingredients in U.S. Licensed Vaccines | FDA](#)

Type of Ingredient	Example(s)	Purpose	Most common source found...
Preservatives	Thimerosal (only in multi-dose vials of flu vaccine)*	To prevent contamination	From eating foods such as certain kinds of fish, mercury (which thimerosal contains) gets into the body
Adjuvants	Aluminum salts	To help boost the body's response to the vaccine	From drinking water, infant formula, or use of health products such as antacids, buffered aspirin, and antiperspirants
Stabilizers	Sugars, gelatin	To keep the vaccine effective after manufactured.	From eating food such as <i>Jell-O®</i> and resides in body naturally
Residual cell culture materials	Egg protein^	To grow enough of the virus or bacteria to make the vaccine	From eating foods containing eggs
Residual inactivating ingredients	Formaldehyde†	To kill viruses or inactivate toxins during the manufacturing process	Resides in body naturally (more in body than vaccines). Also found automobile exhaust, and household furnishing such as carpets and upholstery.
Residual antibiotics	Neomycin	To prevent contamination by bacteria during the vaccine manufacturing process	Antibiotics that people are most likely to be allergic to — like penicillin — aren't used in vaccines

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Vaccine Administration

Before Administration

Proper vaccine storage, staff training and education, screen for contraindications, patient education and vaccine information statement, after care instruction.

Vaccine administration

Infection control (hand hygiene, PPE), vaccine form (nasal spray, single or multidose vial, manufacturer-filled syringe), injection (inject without aspiration) and pain management (use tactile stimulation). Proper waste disposal and proper documentation.

Screening Checklist for Contraindications

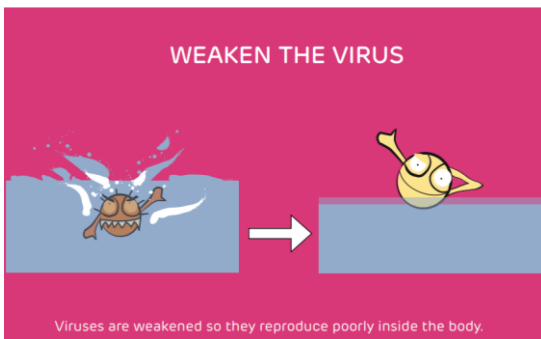
Please ask your patients to provide the following information.

	yes	no	don't know
1. Is the person to be vaccinated sick today?	☐	☐	☐
2. Does the person to be vaccinated have an allergy to an ingredient of the influenza vaccine?	☐	☐	☐
3. Has the person to be vaccinated ever had a serious reaction to influenza vaccine in the past?	☐	☐	☐
4. Is the person to be vaccinated younger than age 2 years or older than age 49 years?	☐	☐	☐
5. Does the person to be vaccinated have a long-term health problem with heart disease, lung disease (including asthma), kidney disease, neurologic disease, liver disease, or metabolic disease (e.g., diabetes)?	☐	☐	☐
6. If the person to be vaccinated is a child age 2 through 4 years, in the past 12 months, has a healthcare provider told you the child had wheezing or asthma?	☐	☐	☐
7. Does the person to be vaccinated have a) an open channel between the cerebrospinal fluid (CSF) and the mouth, throat, nose or ear or any other cranial CSF leak, or b) a cochlear implant, or c) an immunocompromising condition due to any cause (e.g., medication, congenital or acquired immunodeficiency, HIV infection, or a missing or non-functioning spleen [e.g., caused by sickle cell disease])?	☐	☐	☐
8. Is the person to be vaccinated currently taking influenza antiviral medications, or have they taken any within the past 3 weeks?	☐	☐	☐
9. Is the person to be vaccinated a child or teen age 6 months through 17 years and receiving aspirin- or salicylate-containing medicine?	☐	☐	☐
10. Is the person to be vaccinated pregnant or could they become pregnant within the next month?	☐	☐	☐
11. Has the person to be vaccinated ever had Guillain-Barré syndrome?	☐	☐	☐
12. Does the person to be vaccinated live with or expect to have close contact with a person whose immune system is severely compromised and who must be in protective isolation (e.g., an isolation room of a bone marrow transplant unit)?	☐	☐	☐
13. Has the person to be vaccinated received any other vaccinations in the past 4 weeks?	☐	☐	☐

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Living/attenuated & Killed/dead vaccine



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1- Live, attenuated Vaccines

- Weaker version of the virus that can induce protective immunity without virulence.
- Live attenuated vaccines contain the three elements required for a strong immune response:
 - 1- target antigen.
 - 2- PAMPs for stimulation of innate immunity (adjuvant effects).
 - 3- Efficient delivery resulting from the natural invasiveness of the organism.

Immune system will deal with the live attenuated vaccine as the original infectious pathogen.

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Methods to reduce or eliminate virulence:

1. Passaging: Forcing the pathogen to repeatedly replicate in cell culture.
2. Introducing the pathogen into a new species (not the natural host for the pathogen).
3. Attenuation: Growing a pathogenic bacterium or virus for prolonged periods under suboptimal culture conditions. For example, BCG vaccine for vaccination against tuberculosis.
4. Genetic engineering to selectively remove the virulence genes.

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Advantages:

Robust immune response. Therefore, these vaccines are effective after a single immunization dose.

Disadvantages “safety”

- a) Identifying a safe attenuated strain is difficult.
- b) Immunocompromised patients can get sick even with the attenuated virus.
- c) Generally contraindicated in pregnancy.

Examples: bacteria: *Mycobacterium tuberculosis vaccine (BCG)*. Virus: measles-mumps and rubella (MMR), polio and Mpox/smallpox combined vaccine

2- Inactivated or killed organisms

- Depends on inactivating or killing a pathogenic organism.
- To prepare the inactivated vaccine, the living pathogen is inactivated by heat or treatment with chemical agents like formaldehyde or alkylating agents.
- It is important to maintain the structure of epitopes on surface antigens during inactivation. Heat inactivation is often unsatisfactory because it causes extensive denaturation of proteins. Therefore, surface proteins are significantly altered.

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Disadvantages: Killed-pathogen vaccines are generally less immunogenic than live vaccines.

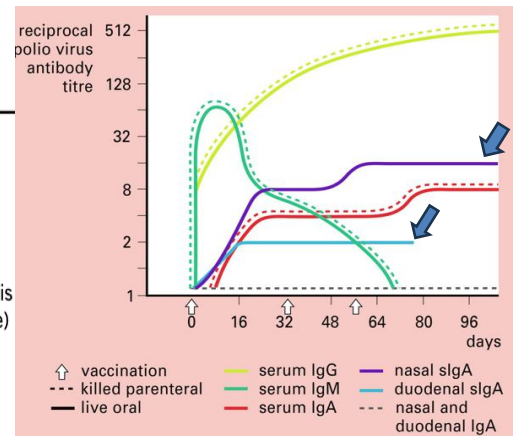
To compensate:

1. Killed-pathogen vaccines are often administered with an adjuvant (such as alum) to increase potency.
2. Killed vaccines often require repeated boosters to achieve the required protective immune status.

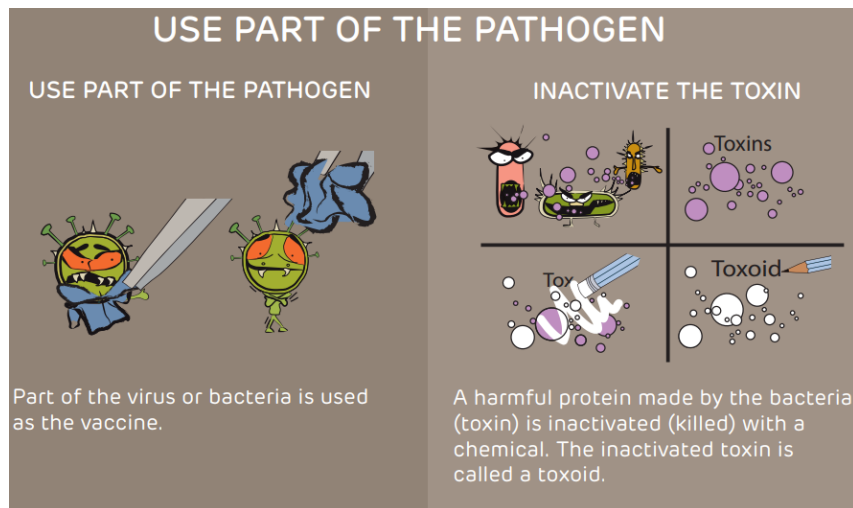
Examples include vaccines against influenza virus, hepatitis A virus, polio virus, and SARS-CoV-2

Compare live and killed vaccines

Property	Live	Dead
Stimulation of an immune response	Effective upon a minimal number of administrations	Requires multiple administrations
Infectious nature of the vaccine	Infectious	Noninfectious
Incidence of adverse reaction to vaccine administration	Can cause infection in immunocompromised individuals	Common for certain vaccine formulations (e.g., pertussis component of DTP vaccine)



3- Purified or recombinant subunit vaccines



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Purified or recombinant subunit vaccines

Administration of a dominant pathogen antigen can efficiently focus the immune response against the immunogenic moiety.

Three common applications:

- a- Inactivated exotoxins or toxoids.
- b- Capsular polysaccharides.
- c- Recombinant protein antigens.

a) Toxoid:

Bacterial toxin is weakened/destroyed by chemicals/heat.

Toxic properties are lost, and the immunogenicity remains, which allows production of antibodies. Toxoids induce active immunization without toxicity.

Examples: *Clostridium tetani* and *Corynebacterium diphtheriae*, produce exotoxins that induces pathogenesis in the infected host.

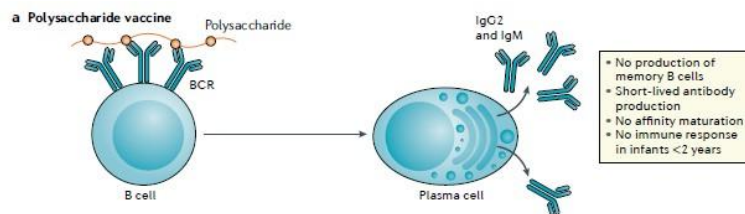
DTaP vaccine = immune response and antibody-mediated neutralization of toxins to prevent disease.

b. Capsular Polysaccharides:

Polysaccharides of certain capsular bacteria (for example, *Neisseria meningitidis* and *Streptococcus pneumoniae*) can elicit an immune response.

Disadvantage

Due to their non-protein nature, polysaccharides are generally unable to activate Th cells. Instead, they activate B cells in a T-independent manner, resulting in antibody production but no affinity maturation.

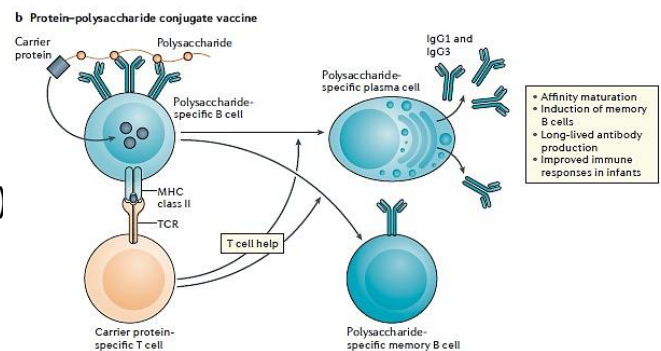


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Way out:

- Conjugate the polysaccharide antigen to a protein carrier to induce T helper cell responses.
- **“Conjugate Vaccine”**. For example, *Haemophilus influenza* type b (Hib) vaccine and *N. meningitis*.



c) Recombinant Protein antigens:

Genes or gene fragments encoding immunogenic protein is cloned and expressed in yeast or bacteria by recombinant DNA techniques.

Examples include hepatitis B vaccine, developed by cloning the gene for the major hepatitis B surface antigen (HBsAg; like Hcpisav-B is administered to adults as 2 doses 1 month apart)

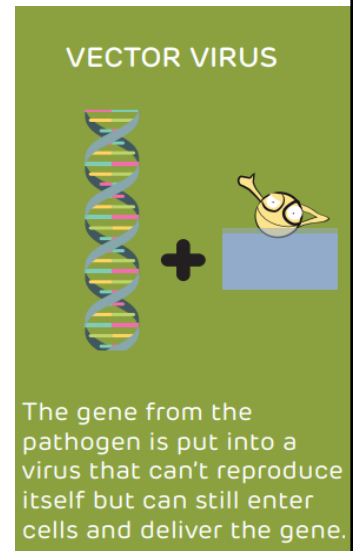
Protective levels of neutralizing antibodies were produced in 95% of vaccinated children, and adequate titers are persistent for 10-18 years.

4. Recombinant vector vaccines

The key antigen of the virulent pathogen is introduced (recombined) into another attenuated viruses or bacteria as vaccinia virus, adenoviruses or attenuated *Salmonella* strains.

After injection, the attenuated organism serves as a vector (carrier) that replicates within the host and generate the protein product of the cloned pathogen.

Recombinant vector vaccines maintain the advantages of live attenuated vaccines while avoiding major disadvantage.



One major challenge for this vaccine: vaccinated individual may mount an immune response against the viral/bacterial vector which can abolish the efficacy of the vaccine especially when a booster shot is needed, **Why?**

In the second or booster shot, the immune system might recognize the viral vector as the enemy and create neutralizing antibodies against it. Therefore, -----

Advantages of Adenovirus vector in vaccines:

- 1- Adenoviruses are easy to manipulate genetically (large-sized and well-characterized genome).
- 2- Adenoviruses cause mild infections in humans and their viral replication can be inhibited by genetic modifications (fewer side-effects).
- 3- Broad tropism, thermostability and ability to grow to high titers. Immunogenicity (PAMPs).
- 5- Minimal risk of integration in the host genome compared to other DNA viruses.

Challenges:

More complicated manufacturing process. Live viruses may evolve to lose the extra genes during vaccine manufacture or during replication within an individual, which may limit the vaccine efficacy.

5. Virus-like particle

Virus-like particles (VLPs) are multiprotein structures that mimic the organization and shape of the virus but lack the viral genetic material. First identified in 1968 in the sera of patients with Down's syndrome, leukemia and hepatitis. VLPs had antigenic sites on their surfaces.

Advantage:

- 1- Potentially safer and cheaper vaccine candidates.
- 2- strong immunogenicity.

Challenges:

Technical difficulties.

- **Examples:**

1- Bivalent HPV vaccine (Cervarix, 2vHPV, type 16 &18).

2- Quadrivalent HPV vaccine (Gardasil, 4vHPV). Gardasil is effective against HPV types 6, 11, 16, and 18.

3- 9-valent HPV vaccine (Gardasil 9, 9vHPV). 9-valent HPV vaccine: non-infectious recombinant vaccine: VLPs of capsid protein of HPV types 6, 11, 16, 18, 31, 33, 45, 52, and 58.

6. DNA vaccines

A plasmid DNA encoding antigenic proteins are injected naked (without viral or bacterial vector) into the recipient.

Following injection, DNA vaccine is engulfed by resident DCs. The encoded antigen is presented using MHCs to activate the humoral and cell mediated immunity.

Advantage: stability (no refrigeration is required for plasmid DNA storage)

Challenges: Difficulty of efficient delivery (no vector). Possibility of DNA integration into the host genome that can induce tumorigenesis.

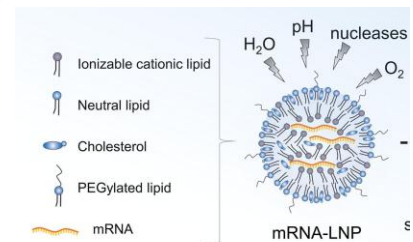
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7. mRNA vaccines

In the cytosol, ribosomes make the protein (antigen). Antigenic protein can be presented on the cell surface (via MHC I and II). Can activate T and B-cell response.

Lipid nanoparticles is a common mRNA delivery tools.

Lipid nanoparticles (LNP) consist of a cationic lipid (positive charged head and hydrophobic tail), which promotes a spherical membranous structure and allows the release of mRNA (negatively charged) to the cytoplasm. This system improves the stability and transfection rates.



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Advantages

1. Safety
2. Efficacy: various modifications make mRNA more stable and highly translatable. Efficient *in-vivo* delivery can be achieved by formulating mRNA into lipid nanoparticles carrier molecules.
3. Production: mRNA vaccines have the potential for rapid, inexpensive and scalable manufacturing.

Challenges:

Stability

Duration of antibody response and ability to self-amplify the response.

General vaccine side effects

- Side effects: injection site pain, redness and swelling and some systemic symptoms such as fever, malaise and headache. Those side effects are attributed to the inflammatory and innate immune response to vaccines. They occur in the first 1–2 days following vaccination.
- Serious side effects from vaccines are rare. Anaphylaxis is most common of these rare side effects. Individuals with known allergies (ex: egg) should avoid vaccines with egg traces.
- CDC and FDA maintain a vaccine safety system named Vaccine Adverse Event Reporting System (VAERS).

[The Vaccine Adverse Event Reporting System \(VAERS\) Request \(cdc.gov\)](#)

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Future directions “Therapeutic vaccines”

- An efficacious therapeutic vaccine is the target.
- Future goal is to create therapeutic vaccines that can stimulate immune responses to eliminate tumors, cure autoimmune diseases, HIV or relieve allergy.

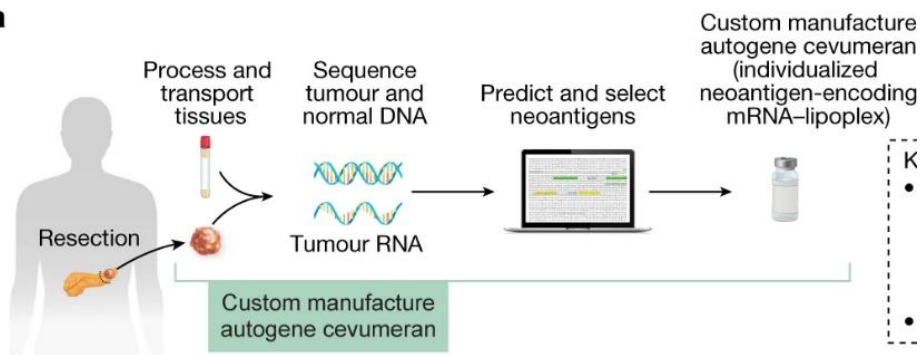
Sipuleucel-T (Provenge®) stimulates the body's immune system to help target and combat prostate cancer cells.

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